Assignment VIII

1. **No-Cloning Theorem**

   We may also prove the No-Cloning Theorem using another method (i.e. by contradiction). We know $\boldsymbol{L}(|a\rangle_{\boldsymbol{A}}|0\rangle_{\boldsymbol{B}}) = |a\rangle_{\boldsymbol{A}}|a\rangle_{\boldsymbol{B}}, \boldsymbol{L}(|b\rangle_{\boldsymbol{A}}|0\rangle_{\boldsymbol{B}}) = |b\rangle_{\boldsymbol{A}}|b\rangle_{\boldsymbol{B}}$ and also $\boldsymbol{L}(\frac{\mathbf{1}}{\sqrt{\mathbf{2}}}(|a\rangle_{\boldsymbol{A}} + |b\rangle_{\boldsymbol{A}})|0\rangle_{\boldsymbol{B}}) = \frac{\mathbf{1}}{\sqrt{\mathbf{2}}}(|a\rangle_{\boldsymbol{A}} + |b\rangle_{\boldsymbol{A}})\frac{\mathbf{1}}{\sqrt{\mathbf{2}}}(|a\rangle_{\boldsymbol{B}} + |b\rangle_{\boldsymbol{B}})$ by definition, where $|a\rangle$ and $|b\rangle$ are arbitrary normalized states. Expand the last equation and evaluate $\boldsymbol{L}(|a\rangle_{\boldsymbol{A}}|0\rangle_{\boldsymbol{B}} + |b\rangle_{\boldsymbol{A}}|0\rangle_{\boldsymbol{B}})$ to show that there is contradiction (that |a⟩ and |b⟩ cannot be generalized states) and thus cloning of a general state is impossible.

2. **Quantum Teleportation of a Superposition State**

   a) Figure 1 shows the simplified version of quantum teleportation. The state to be teleported is |+⟩. Explain why this is the state to be teleported?

   b) Construct the matrix (4 × 4) from the left to Alice's first measurement.

   c) Explain the simulation results and why quantum teleportation is successful.

   d) Note that the MSB contains not just 0 but also 1 in the final output. Why the two Hadamard gates do not cancel each other? Hints: Look at the matrix constructed.

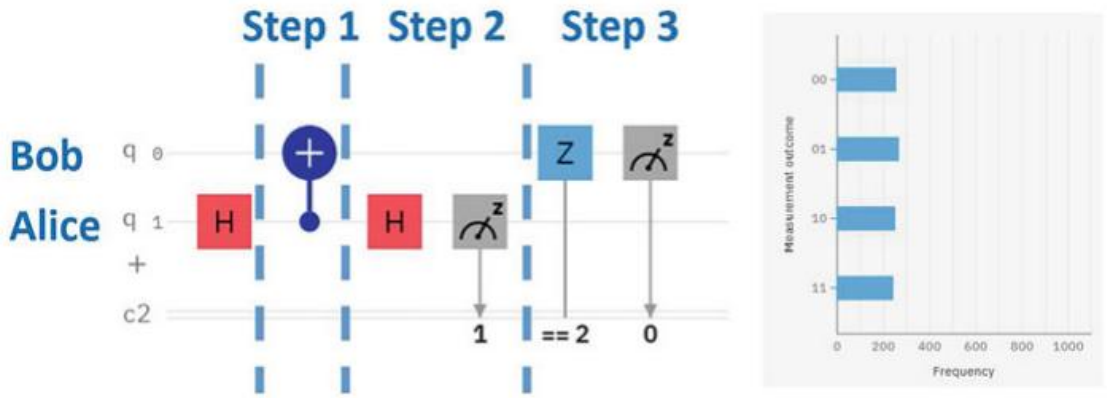


Figure 1 Quantum teleportation circuit corresponds to Pic. 20 in Chapter 18 with |Ψ⟩ = |0⟩

3. **No-Cloning Theorem and Heisenberg Uncertainty Principle**

   If the state is a basis state, it is possible to clone. For any arbitrary state, it must be a basis state of another basis. Can we clone it? How is it related to the Heisenberg Uncertainty Principle? Please discuss.

4. Entanglement

Derive the following equation using matrix.

$$\frac{1}{\sqrt{2}}(\alpha|000\rangle + \beta|110\rangle + \alpha|011\rangle + \beta|101\rangle) = \frac{1}{\sqrt{2}}(\alpha|000\rangle + \beta|101\rangle + \alpha|011\rangle + \beta|110\rangle)$$

5. Classical Register and Conditional Gate

a) Reproduce the result in Figs. 2 and 3.

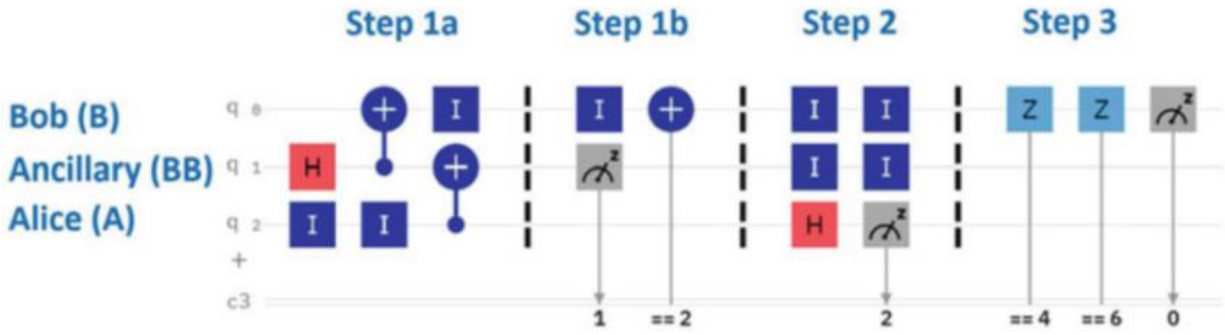


Figure 2 Implementation of the full quantum teleportation algorithm

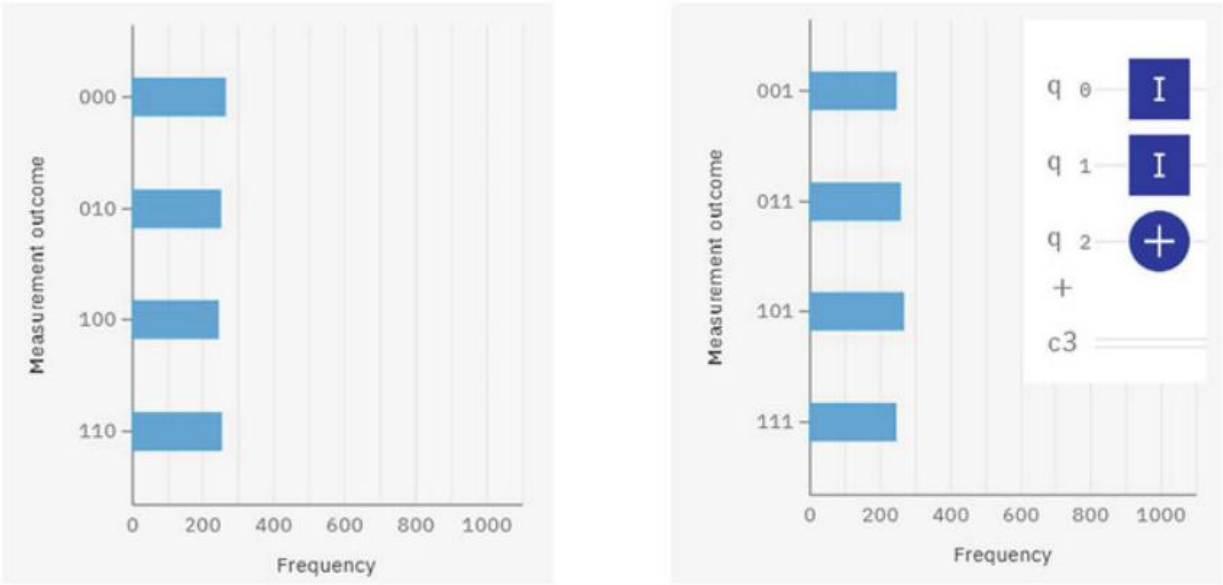


Figure 3 Simulation results of the circuit in Fig. 1 with |Ψ⟩ = |0⟩ (left) and |Ψ⟩ = |1⟩ (right). The circuit to create |Ψ⟩ = |1⟩ is shown as the inset

b) Store the Step 1b measurement result in the 0th classical qubit and store Step 3 measurement result in the 1st classical qubit. How should we modify the conditional CNOT and Z-gates in Step 1b and Step 3 so it will produce the same result? Hints: The classical bit we want to inspect now is the 1st qubit.

6. Quantum Teleportation vs. Hypothetical Teleportation

Go through the quantum teleportation one more time and answer the following questions: (1) Why the state of the original electron is destroyed? How is it related to the no-cloning theorem? (2) If you want to quantum teleport myself to Moon or Mars and back to the Earth, how should you prepare for that? (3) If Alice's state was $|0\rangle$, do we need quantum teleportation to transfer the state to Bob? Do we violate the no-cloning theorem?

7. Other Forms of $\boldsymbol{f(x)}$

Here is what we did in f_A as below:

$$\boldsymbol{U}_f|0\rangle|0\rangle = |0\rangle|0 \oplus f(0)\rangle = |0\rangle|0 \oplus 0\rangle = |0\rangle|0\rangle$$

$$\boldsymbol{U}_f|0\rangle|1\rangle = |0\rangle|1 \oplus f(0)\rangle = |0\rangle|1 \oplus 0\rangle = |0\rangle|1\rangle$$

$$\boldsymbol{U}_f|1\rangle|0\rangle = |1\rangle|0 \oplus f(1)\rangle = |1\rangle|0 \oplus 0\rangle = |1\rangle|0\rangle$$

$$\boldsymbol{U}_f|1\rangle|1\rangle = |1\rangle|1 \oplus f(1)\rangle = |1\rangle|1 \oplus 0\rangle = |1\rangle|1\rangle$$

where,

$$f_A(0) = 0 \text{ and } f_A(1) = 0$$

$$f_B(0) = 0 \text{ and } f_B(1) = 1$$

$$f_C(0) = 1 \text{ and } f_C(1) = 0$$

$$f_D(0) = 1 \text{ and } f_D(1) = 1$$

Repeat what we did in f_A for f_B, f_C, and f_D.

8. Creation of Superposition

Repeat Eq. $(\boldsymbol{HH})(\boldsymbol{IX})|0\rangle|0\rangle = (\boldsymbol{HH})|0\rangle|1\rangle = |+\rangle|-\rangle$ using matrix.

9. Hadamard Gate Applied to 2-Qubit Basis State

Creating the superposition in Step 1 in Fig. 1 after the NOT-gate is equivalent to applying a 2-qubit Hadamard gate to state |01⟩. Show that you can get the result in the below equation by using Eq. $\boldsymbol{H^{\otimes n}|y\rangle = \frac{1}{2^{n/2}}\sum_{x=0}^{2^n-1}(-1)^{x.y}|x\rangle}$.

$$|+\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$= \tfrac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

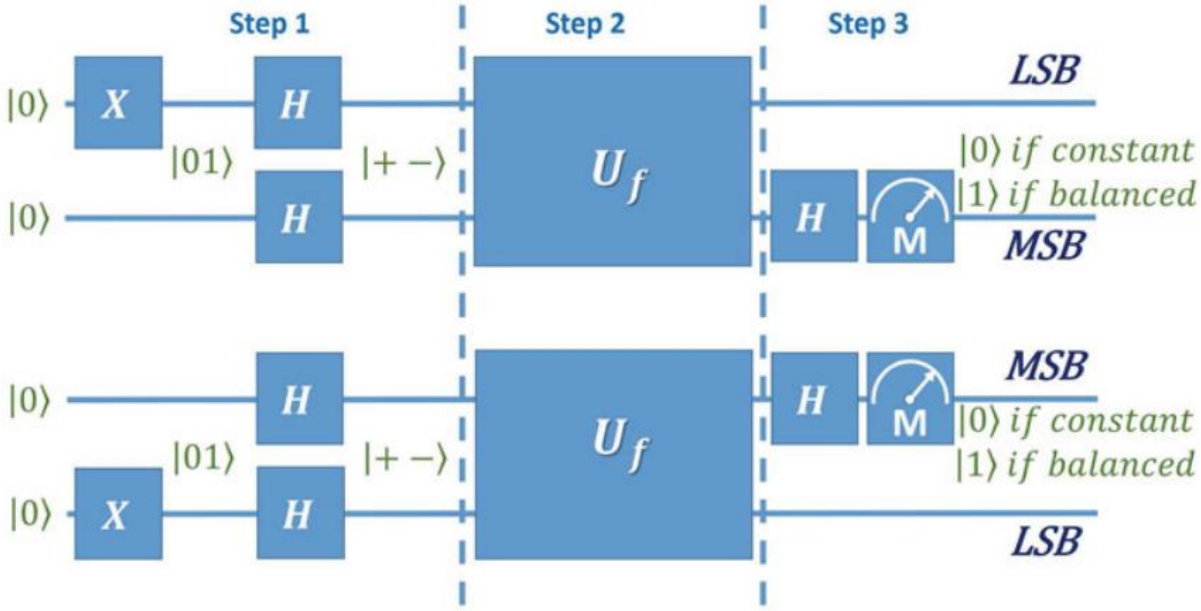


Figure 1 The Deutsch algorithm circuit. The top is the convention we have been using (IBM-Q) with MSB at the bottom and the bottom one is the other version often seen in other books. Both are using little-endian convention

10. Non-Constant Function in the Deutsch algorithm

Repeat the following example for a balanced function. Note that in the example, f(x) is a constant function that always gives "0".

$$\boldsymbol{U_f}|0\rangle|0\rangle = |0\rangle|0 \oplus f(0)\rangle = |0\rangle|0 \oplus 0\rangle = |0\rangle|0\rangle$$

$$\boldsymbol{U_f}|0\rangle|1\rangle = |0\rangle|1 \oplus f(0)\rangle = |0\rangle|1 \oplus 0\rangle = |0\rangle|1\rangle$$

$$\boldsymbol{U_f}|1\rangle|0\rangle = |1\rangle|0 \oplus f(1)\rangle = |1\rangle|0 \oplus 0\rangle = |1\rangle|0\rangle$$

$$\boldsymbol{U_f}|1\rangle|1\rangle = |1\rangle|1 \oplus f(1)\rangle = |1\rangle|1 \oplus 0\rangle = |1\rangle|1\rangle$$